BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Wired and Wireless Networks

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. What happens if the central switch of a star network fails?

2. More users share a limited network link. What may happen?

A) More bandwidth per user
B) Less bandwidth per user
C) Each device gains RAM
D) All delays disappear

3. What distinguishes a WAN?

4. Convert denary 150 into eight-bit unsigned binary.

Today's objectives

Distinguish copper, fibre-optic and wireless connections. Explain suitable advantages and drawbacks for each.

Key vocabulary - Read, cover, recall

Wired connection
A link using a physical cable.

Copper cable
Transmission medium carrying electrical signals.

Fibre optic
Transmission medium carrying signals as light.

Wireless connection
Communication without a connecting cable, often radio.

Write one key term and its meaning in your own words:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New learning - Key ideas

Copper cables	Wired copper connections carry electrical signals and are affordable, but have speed/distance limits.
Fibre-optic cables	Fibre carries pulses of light, supporting high rates and long distances; installation can be more costly.
Wireless connections	Wi-Fi uses radio waves so devices can move without cables, but connections may experience interference and interception.
Choose a connection	Use the scenario: high bandwidth, cost, mobility, length of cable, interference and security matter.
Ethernet, Wi-Fi and Bluetooth	Ethernet is a wired networking connection. Wi-Fi uses radio and supports mobility but may suffer interference. Bluetooth suits short-range links such as a nearby keyboard or headphones, considering power and bandwidth. Choose connections from the actual range, mobility and throughput requirements.

I Do - One worked example

A college connects two buildings a long distance apart. Explain why fibre optic may be more suitable than copper. [4 marks]

1. The connection is between buildings over a substantial distance.
2. Fibre uses light pulses [1] and is suitable for high-speed, longer-distance links [1].
3. Copper carries electrical signals [1] and has greater limitations on speed/distance for such a link [1].
4. Two contrasts, each explained in context, make four points.

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A cafe provides wireless access to customers using phones and tablets. Explain two reasons Wi-Fi is suitable. [4 marks]

2. A business handles confidential data over Wi-Fi. Explain two risks and how encryption helps. [4 marks]

3. A hospital must connect fixed machines reliably. Explain one benefit and one drawback of a wired connection. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Predict the ideal transfer time in seconds.

```
fileMb = 120
speedMbps = 60
print(fileMb DIV speedMbps)
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State one disadvantage of wireless transmission. [1 mark]

Q3. A research station must connect an outdoor field laboratory to its main building. Explain one advantage and one limitation of using fibre-optic cable for this long link. [4 marks]

Q4. Compare copper, fibre and wireless communication for a school. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A wireless connection always needs a separate cable from every phone to the access point. ☐ T ☐ F

2. Wired copper connections carry electrical signals and are affordable, but have speed/distance limits. ☐ T ☐ F

3. Fibre carries pulses of light, supporting high rates and long distances; installation can be more costly. ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 If the central switch fails, devices using it lose contact. A broken cable usually affects just one workstation. ☐

Q2 B - Less bandwidth per user ☐

Q3 A WAN links networks over a large area, often using links provided by other organisations. ☐

Q4 (a) 10010110. ☐

Code First - Model

```
fileMb = 120
speedMbps = 60
print(fileMb DIV speedMbps)
```

Output: 2

We Do - Check the model

1. Wireless uses radio rather than cables to each device [1], allowing users to move while connected [1]. Devices can join without dedicated physical ports [1], making guest access more flexible [1]. ☐

2. Wireless signals can be intercepted by nearby devices [1], creating a confidentiality risk [1]. Encryption scrambles transmitted data [1], so captured content cannot be understood without the key [1]. ☐

3. Wired cables provide a stable link less affected by radio interference [1], improving reliability [1]. The cable restricts device movement [1], reducing flexibility if machines are relocated [1]. ☐

Exit Ticket - Check

1. FALSE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 Interference, interception/security risk or variable signal quality [1]. ☐

Q3 Fibre carries light-based signals [1] and is well suited to high-speed/long-distance links [1]. Installing fibre can be more costly or difficult [1], increasing the cost/complexity of the research station's connection [1]. ☐

Q4 Copper carries electrical signals [1] and is cost-effective for local wired links [1]. Fibre carries light [1] and offers high speed/long distance [1]. Wireless uses radio [1] and offers mobility but may suffer interference/interception [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: